

Emerging Data Citation Standards

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VALUE OF PROPER CITATION

- Proper citations make it easier to **SEARCH FOR EXISTING DATA AND TOOLS** and users are more likely to give credit with the citation method is clear .
- Researchers can **MEASURE IMPACT** by tracking DOIs across multiple contexts:
 - (1) scientific research
 - (2) the media
 - (3) on social media
- Proper citations enable **REPRODUCIBLE WORKFLOWS**, and thus promote open science best practices
 - Reproducible pipelines make analytics **EXTENSIBLE**, meaning new insights can be generated **FASTER AND MORE RELIABLY** using existing infrastructure and workflows
 - It is easier to identify and **CORRECT SCIENTIFIC ERRORS** in research

CHALLENGES OF DATA CITATION SYSTEMS

- They are not well-established standards for citing data
 - Data is typically generated as part of a research project and has historically been references through published research done with the data, not as a distinct citation.
- Data is not one thing: is can be tables, text, crosswalks, images, video, voice, or workflows
- Unlike publications, “data” looks more like a Russian Doll than a single entity:
 - A panel includes many years of data
 - A database includes many tables
 - A data series can include multiple databases
 - An ecosystem can link multiple data series
- Data is dynamic:
 - Panels are updated by adding new waves
 - Data is cleaned and refined
 - Linking datasets can add new columns to existing data
- Standards for defining “distinct” or enhanced are poorly-defined:
 - Do modifications to an existing dataset constitute new data?
 - Does linking two datasets result in a new dataset?

DIGITAL OBJECT IDENTIFIERS (DOIs)

DOI STRUCTURE

<https://doi.org/10.5555/YRFU1371>



prefix = organization

suffix = object id (ex. pub, data)

In the DOI database the citations can be hierarchical:

<https://doi.org/10.7910/DVN/1EMHTK>

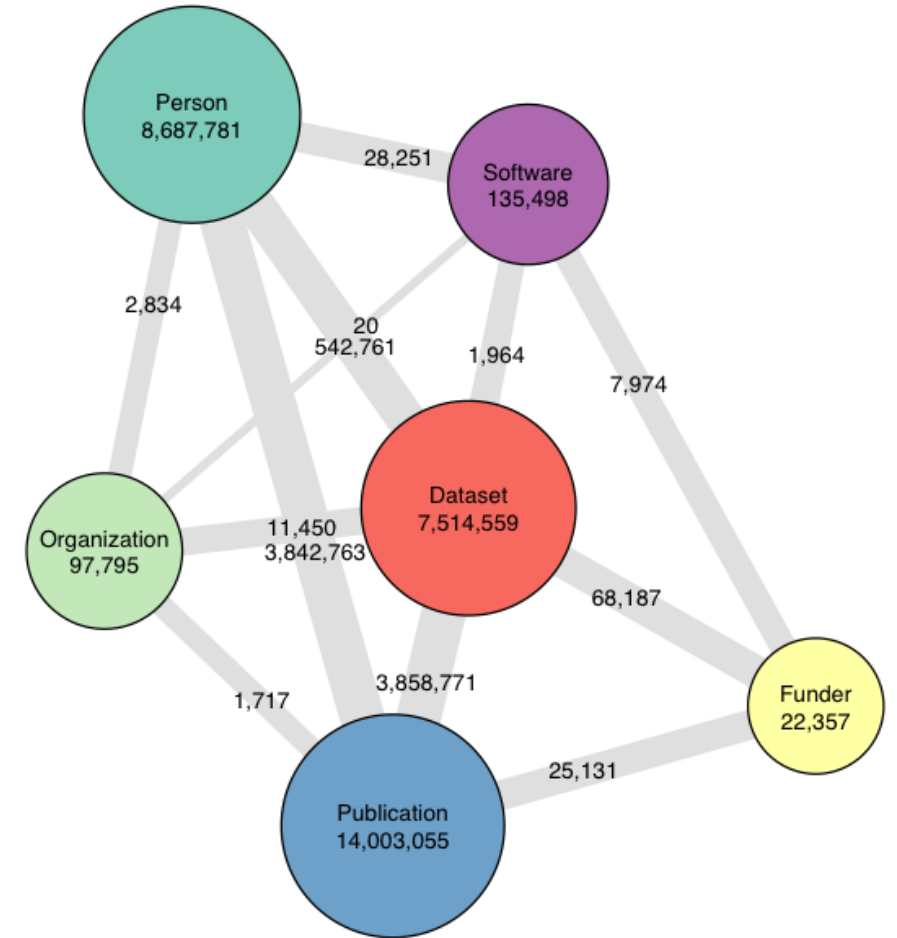
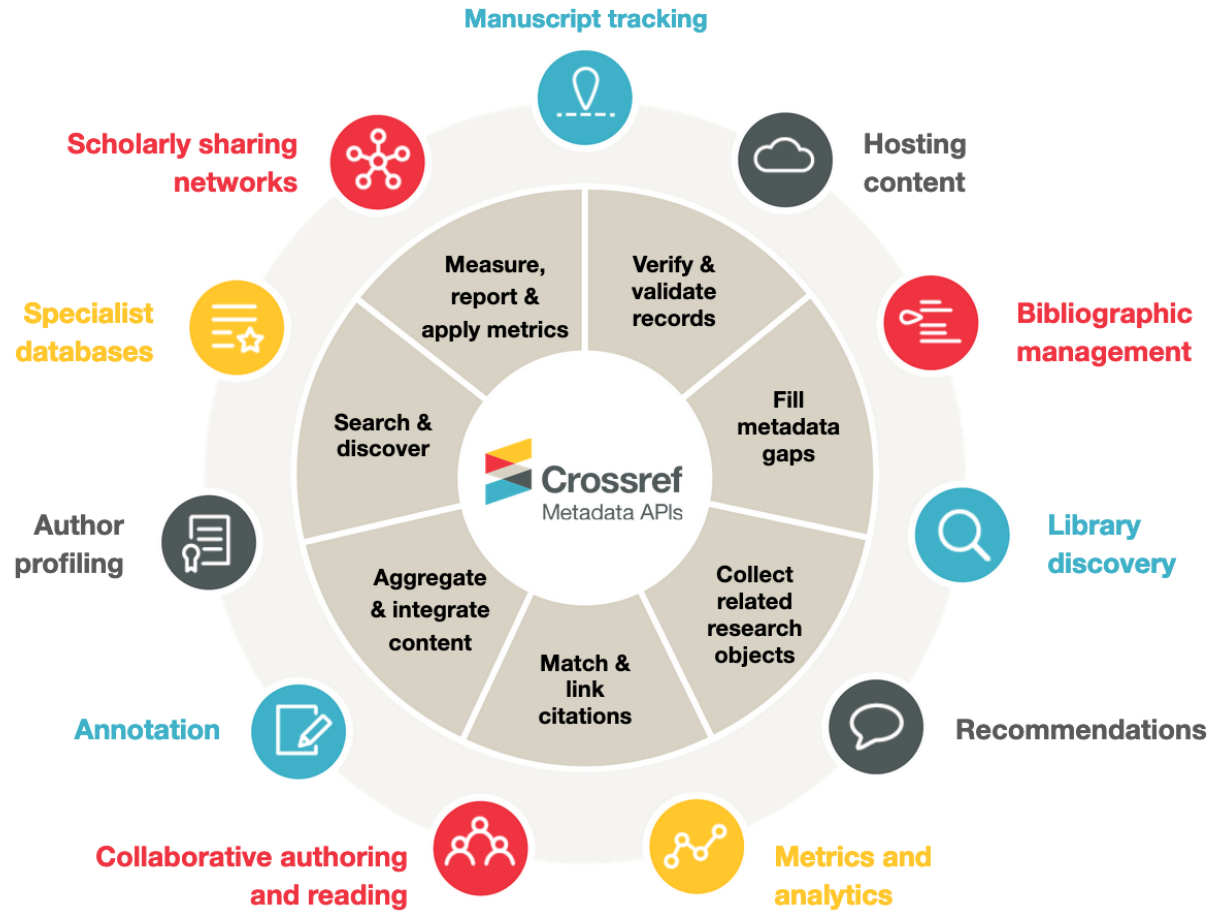
Harvard Dataverse project ID object ID

Example:

Gary King, Benjamin Schneer, & Ariel White. (2017). Replication Data for: How the News Media Activates Public Expression and Influences National Agendas. Harvard Dataverse.

<https://doi.org/10.7910/DVN/1EMHTK>.

VALUE OF DOI DATABASE:



The Altmetric Attention Score

The colors of the Altmetric donut each represent a different source of attention:



**DOIs Enable Tracking
Content Use –
The First Step of
Measuring Impact**

The amount of each color in the donut will change depending on which sources a research output has received attention from:



This output has received a lot of mainstream media coverage

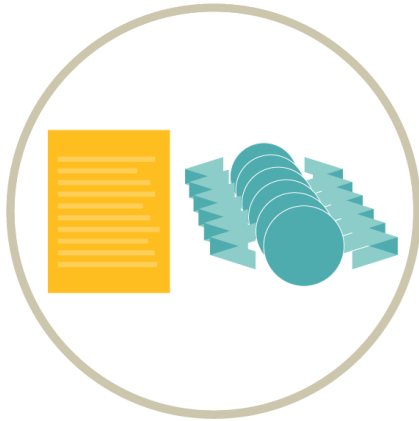


This research has received most of its attention from blogs and has been referenced in public policy documents

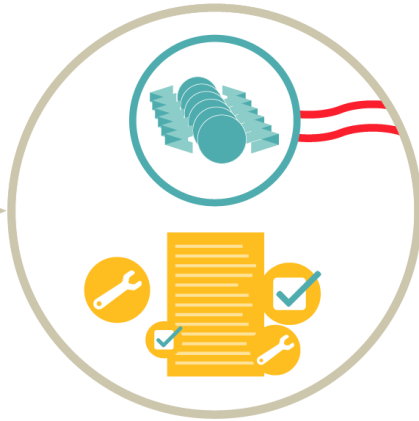


This research has received a lot of attention on Twitter and has also been reviewed on a post-publication peer-review forum

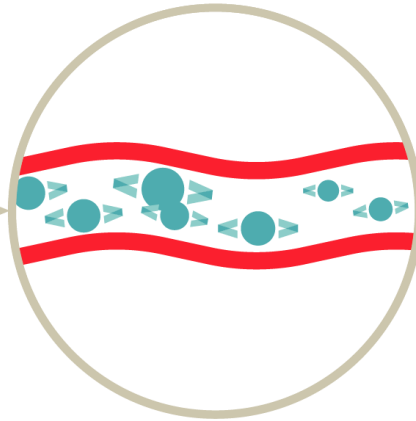
Metadata Retrieval: how it works



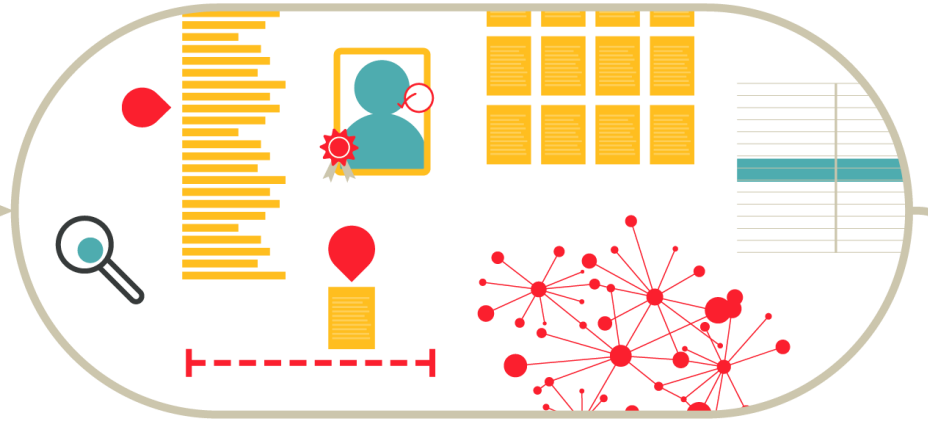
Publisher members send us information called metadata that we collect and store in a standard way.



We make the metadata openly available via APIs so that people and computers can incorporate it in their tools and services.



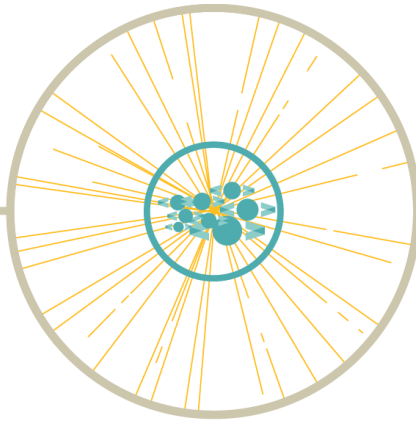
Our REST API helps systems to get the most complete, up-to-date set of metadata from all our member publishers.



Search services, manuscript tracking systems, bibliographic management software, library systems, author profiling tools, specialist subject databases, and scholarly sharing networks all use our metadata.



Researchers benefit from a service that helps support, progress, publish, and share their work.



Developers can then come to one place to find and use information from tens of millions of publications.



This means publishers only have to send metadata to one place to have their content discovered and used by many different services.



This helps their users find, link to, cite, and assess the research they are interested in.

CUSTOMIZING DOIs

POSSIBLE NCCS DOIs

doi/xx-urban/	pub/report/id
doi/xx-urban/	pub/book/id
doi/xx-urban/	pub/blog/id
doi/xx-urban/	pub/series/id
doi/xx-urban/	data/meta/npname
doi/xx-urban/	data/meta/nplocations
doi/xx-urban/	data/crosswalk/ntee-to-pcs

PUBLICATIONS SERIES:

pub/sectorbrief/sib2018
pub/sectorbrief/sib2019

DATA SERIES:

data/core
data/bmf
data/pub78

TABLES IN A DATABASE?

data/meta/npname
data/meta/nplocations
data/meta/npmissions

**HOW DO YOU
CITE DATA?**

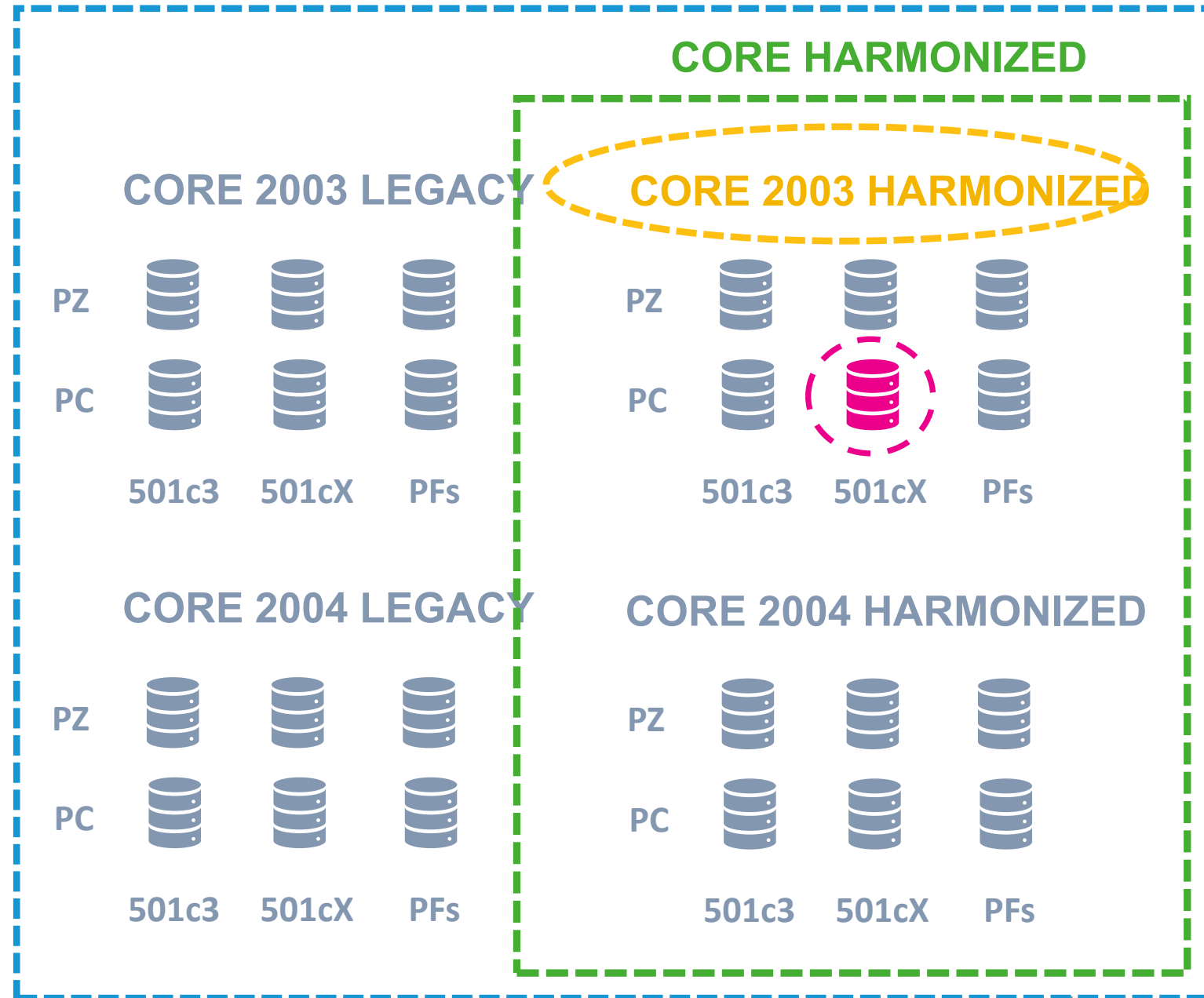
Should you cite a data series, database, or table?

Data has time dimensions (annual waves), different tables for different orgs, and multiple versions.

- NCCS Data, downloaded in 2024 (typical, but not helpful)
- NCCS Core: [doi/xxx](#)
- NCCS Core: [doi/xxx/V3.41](#)
- NCCS Core: [doi/xxx/V3.41 \(2003\)](#)
- NCCS Core: [doi/xxx/V3.41/PC-501cX-2003](#)

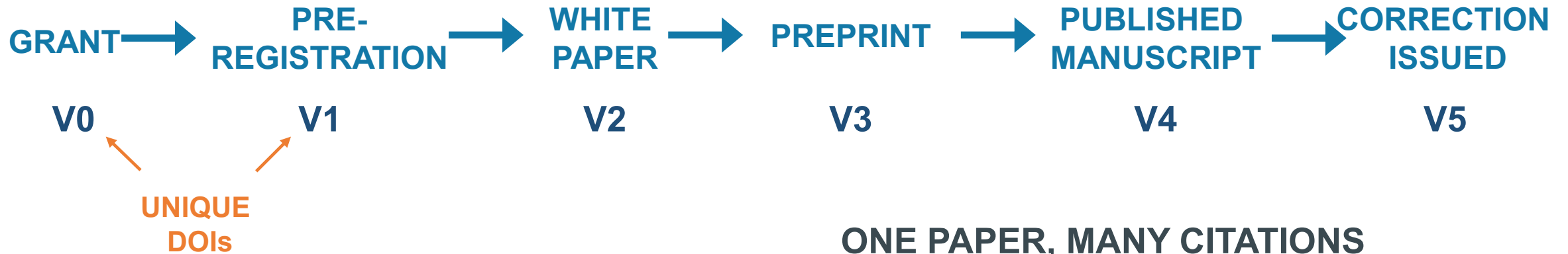
ONLY THE LAST VERSION
IS REPRODUCIBLE IN A
WORKFLOW

EXAMPLE USING NCCS CORE DATA SERIES



LIFE-CYCLE OF A RESEARCH PAPER THROUGH DOIs

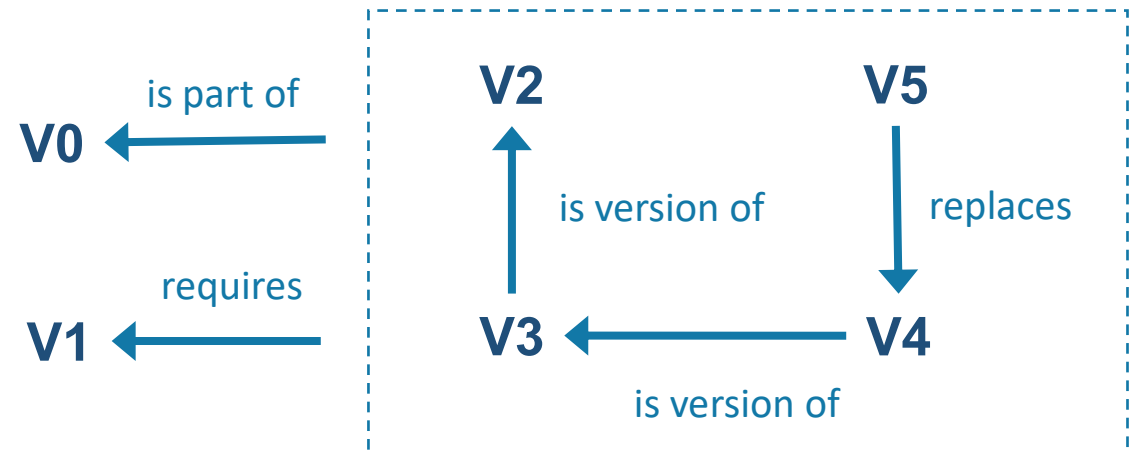
Research has a life-cycle and products takes on different forms at different stages.



DOI DATABASES USE DUBLIN CORE ATTRIBUTES TO LINK DOIs

- Is Version Of
- Has Version
- Is Replaced By
- Replaces
- Is Required By
- Requires
- Is Part Of
- Has Part
- Is Referenced By
- References
- Is Format Of
- Has Format
- Conforms To

ONE PAPER, MANY CITATIONS



DATA PROVENANCE (HISTORY OF THE DATASET)

Similarly, data has a life-cycle and takes on different forms at different stages.



Unique DOIs through versioning (ideal case)

UNIQUE DOIs	→	V0:	doi/xxx/ v0.0.0	archive of public dataset
	→	V1:	doi/xxx/ v1.0.0	transformation into new dataset
	→	V2:	doi/xxx/ v1.1.0	improvement (link with other data)
		V3:	doi/xxx/ v1.1.1	fix and error
		V4:	doi/xxx/ v1.2.1	improvement (add new wave)
		V5:	doi/xxx/ v1.2.1/puf	public use version of restricted file

DATA PROVENANCE MEANS NOT TREATING ALL VERSIONS AS THE SAME DATASET

EMERGING CITATION STANDARDS

JOINT DECLARATION OF DATA CITATION PRINCIPLE

1. IMPORTANCE

Data should be considered legitimate, citable products of research. Data citations should be accorded the same importance in the scholarly record as citations of other research objects, such as publications[1].

2. CREDIT AND ATTRIBUTION

Data citations should facilitate giving scholarly credit and normative and legal attribution to all contributors to the data, recognizing that a single style or mechanism of attribution may not be applicable to all data[2].

3. EVIDENCE

In scholarly literature, whenever and wherever a claim relies upon data, the corresponding data should be cited[3].

4. UNIQUE IDENTIFICATION

A data citation should include a persistent method for identification that is machine actionable, globally unique, and widely used by a community[4].

5. ACCESS

Data citations should facilitate access to the data themselves and to such associated metadata, documentation, code, and other materials, as are necessary for both humans and machines to make informed use of the referenced data[5].

6. PERSISTENCE

Unique identifiers, and metadata describing the data, and its disposition, should persist — even beyond the lifespan of the data they describe[6].

7. SPECIFICITY AND VERIFIABILITY

Data citations should facilitate identification of, access to, and verification of the specific data that support a claim. Citations or citation metadata should include information about provenance and fixity sufficient to facilitate verifying that the specific timeslice, version and/or granular portion of data retrieved subsequently is the same as was originally cited[7].

8. INTEROPERABILITY AND FLEXIBILITY

Data citation methods should be sufficiently flexible to accommodate the variant practices among communities, but should not differ so much that they compromise interoperability of data citation practices across communities[8].

Data Citation with Dataverse

Integrates with DataCite service for minting persistent identifiers (DOIs) and registering dataset citation metadata



Authors, Year, Dataset Title, DOI, Data Repository, UNF, version

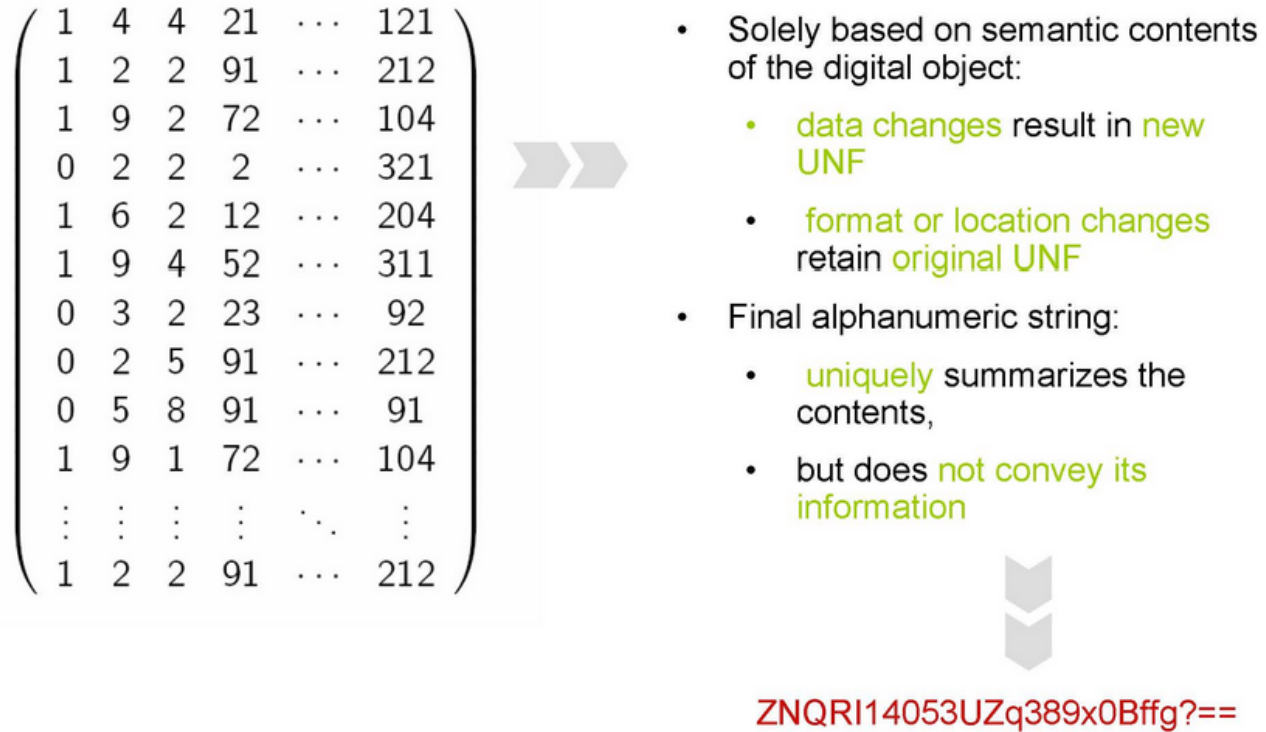


Attribution to data
authors and distributors



Fingerprint (UNF) to verify
dataset, and version to
specify what data are being
referenced. UNF does **not**
depend on the data format.

Universal Numeric Fingerprint (UNF)



1	4	4	21	...	121
1	2	2	91	...	212
1	9	2	72	...	104
0	2	2	2	...	321
1	6	2	12	...	204
1	9	4	52	...	311
0	3	2	23	...	92
0	2	5	91	...	212
0	5	8	91	...	91
1	9	1	72	...	104
⋮	⋮	⋮	⋮	⋮	⋮
1	2	2	91	...	212

- Apply a **cryptographic algorithm**
- Solely based on semantic contents of the digital object:
 - **data changes** result in **new UNF**
 - **format or location changes** retain **original UNF**
- Final alphanumeric string:
 - **uniquely** summarizes the contents,
 - but does **not convey its information**

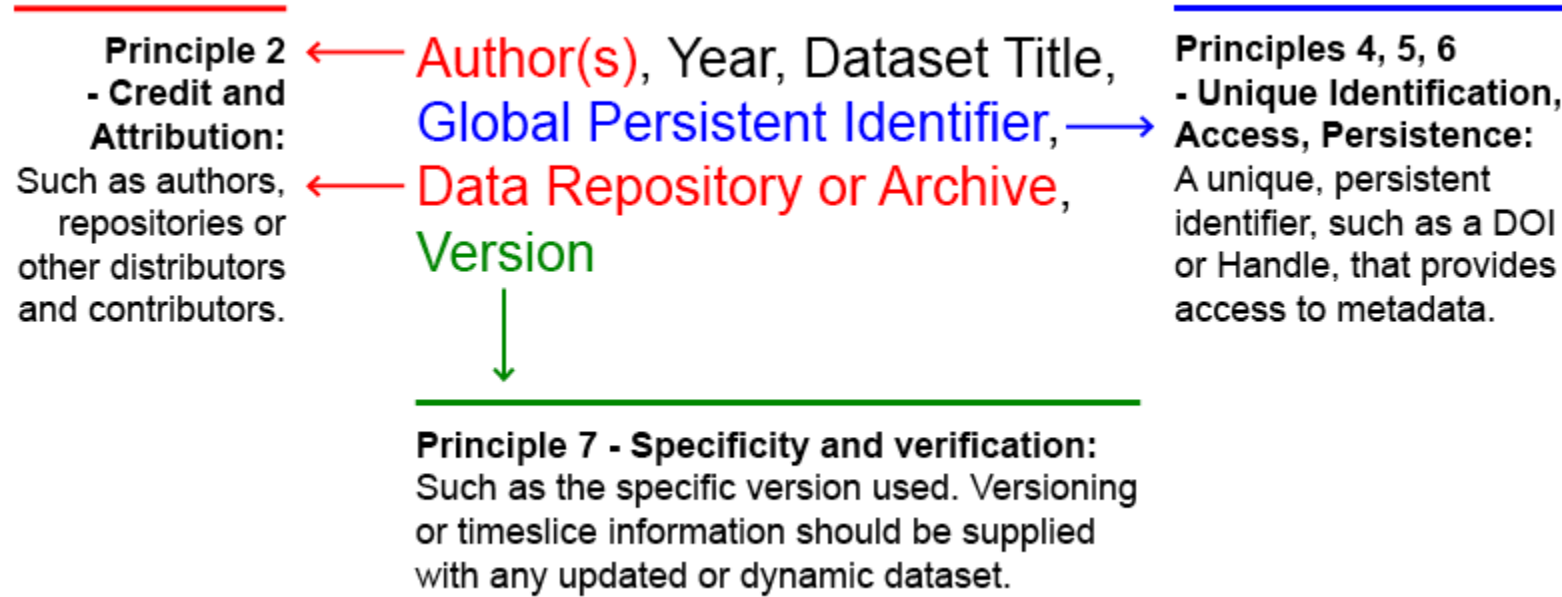
ZNQRI14053UZq389x0Bffg?==

<https://guides.dataverse.org/en/latest/developers/unf/index.html>

Universal Numerical Fingerprint (UNF) is a unique signature of the **semantic content** of a digital object. It is **not** simply a checksum of a binary data file. Instead, the UNF algorithm approximates and normalizes the data stored within. A cryptographic hash of that normalized (or canonicalized) representation is then computed. The signature is thus independent of the storage format. E.g., the same data object stored in, say, SPSS and Stata, will have the same UNF.

Early versions of the Dataverse Software were using the first released implementation of the UNF algorithm (v.3, implemented in R). Starting with Dataverse Software 2.0 and throughout the 3.* lifecycle, UNF v.5 (implemented in Java) was used. Dataverse Software 4.0 uses the latest release, UNF v.6. Two parallel implementation, in R and Java, will be available, for cross-validation.

Learn more: Micah Altman and Gary King. 2007. "A Proposed Standard for the Scholarly Citation of Quantitative Data." D-Lib Magazine, 13. Publisher's Version Copy at <https://j.mp/2ovSzoT>



Illustrated in the figure below, is an example of how the data citation is formulated in a Dataverse repository, using the [Joint Declaration of Data Citation Principles \(2014\)](#) : a synthesis of all previously existing principles and initiatives on data citation.

SEMANTIC VERSION CONTROL

Semantic Versioning Convention

V1.0.0

V1.0.1 (small update, typically bug fix in software)

V1.1.0 (larger update, often a new feature)

V2.0.0 (major update)

Note, if error is corrected the version changes as follows:

periods are categories, not decimals

CORRECT: V1.0.12 --> V1.1.12

INCORRECT: V1.0.12 --> V1.1.00

IN THEORY, categories can be defined for a specific context (data contexts versus software categories):

major version | new wave in panel | data improvement

BMF/v1.03.01

BMF/v2.00.00 # harmonized

BMF/v2.00.01 # NTEE improved, new field added

BMF/v2.01.01 # new month of data integrated

Generating custom DOIs via Crossref:

- In practice, it is currently hard to customize DOIs through Crossref because they don't yet have a form for depositing new datasets – the XML has to be generated manually, including dataset metadata.
- <https://www.crossref.org/documentation/schema-library/markup-guide-record-types/datasets/>

```
<doi_batch xmlns="http://www.crossref.org/schema/4.3.7" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" version="4.3.7" xsi:schemaLocation="http://www.crossref.org/schema/4.3.7 http://www.crossref.org/schemas/crossref4.3.7.xsd">
  <head>
    <doi_batch_id>2006-03-24-21-57-31-10023</doi_batch_id>
    <timestamp>20060324215731</timestamp>
    <depositor>
      <depositor_name>Sample Master</depositor_name>
      <email_address>support@crossref.org</email_address>
    </depositor>
    <registrant>CrossRef</registrant>
  </head>
  <body>
    <database>
      <database_metadata language="en">
        <titles>
          <title>NURSA Datasets</title>
        </titles>
        <institution>
          <institution_name>Nuclear Receptor Signaling Atlas</institution_name>
          <institution_acronym>NURSA</institution_acronym>
        </institution>
        <doi_data>
          <doi>10.1621/NURSA_dataset_home</doi>
          <resource>http://www.nursa.org/template.cfm?threadId=10222</resource>
        </doi_data>
      </database_metadata>
      <dataset dataset_type="collection">
        <contributors>
          <person_name contributor_role="author" sequence="first">
```

<https://gitlab.com/crossref/schema/-/blob/master/best-practice-examples/dataset.5.3.0.xml>

**COMBINING
ALL THREE**

Versions can be incorporated into DOIs

Gary King, Benjamin Schneer, & Ariel White. (2017). Replication Data for: How the News Media Activates Public Expression and Influences National Agendas. Harvard Dataverse. <https://doi.org/10.7910/DVN/1EMHTK>,
version:sha256:0f81072b859159923abd0c47 (“fingerprint” convention using unique data hash)

Gary King, Benjamin Schneer, & Ariel White. (2017). Replication Data for: How the News Media Activates Public Expression and Influences National Agendas. Harvard Dataverse.
<https://doi.org/10.7910/DVN/1EMHTK/v2.15> (versioning convention)

(Zenodo generates a completely unique DOI for each new version)

DOI databases contain metadata elements that link versions:

```
<!--      From Dublin core  -->
```

```
< xsd:enumeration value="isVersionOf" />
```

Versions represent a project or group

Since there is only one DOI created for each dataset and each DOI points to a single URL, a DOI with or without the version number would point to a dataset landing page or changelog that lists all available versions.

`https://doi.org/10.7910/DVN/1EMHTK/v2.15` --> v2.15

`https://doi.org/10.7910/DVN/1EMHTK/` --> landing page with all versions

Semantic Versioning Convention

Currently supported on Dataverse:

major revision / minor revision

BMF/v1.12

Would be better to define custom semantics, but would require Datacite membership:

major version / new wave in panel / data improvement

BMF/v1.03.01

BMF/v2.00.00 # harmonized

BMF/v2.00.01 # NTEE improved, new field added

BMF/v2.01.01 # new month of data integrated

HARVARD DATAVERSE

Harvard Dataverse

Open Science repository for storing data, data workflows, and replication files.

Free for all researchers to use.

Limits:

- 2GB per file

- 1 TB per dataverse (organization or scholar)
(some limits are negotiable)

Advantages:

- R and Python packages make depositing easy

- Supports public and private projects (can use API to access private data)

- Supports versioning

- Has an API for accessing data

Vocabulary:

Dataverse – collection of datasets owned by a person or organization

Dataset – a collection of files

- tabular data
- documentation
- code

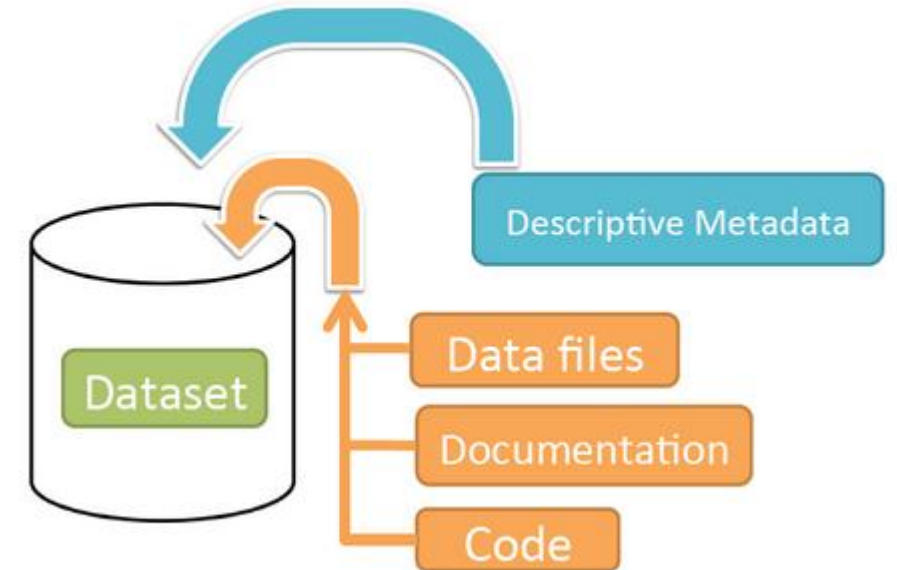
Metadata – many standards supported

<https://guides.dataverse.org/en/4.6/user/appendix.html#metadata-references>

File upload limits

- Maximum of 1,000 files per upload.
- File upload limit is 2.5 GB per file.
- Tabular file ingest is limited to 143.1 MB.

Schematic Diagram of a **Dataset** in Dataverse 4.0



Container for your data, documentation, and code.

See also Data Documentation Initiative (DDI) standards for dataset metadata, and the associated R package:

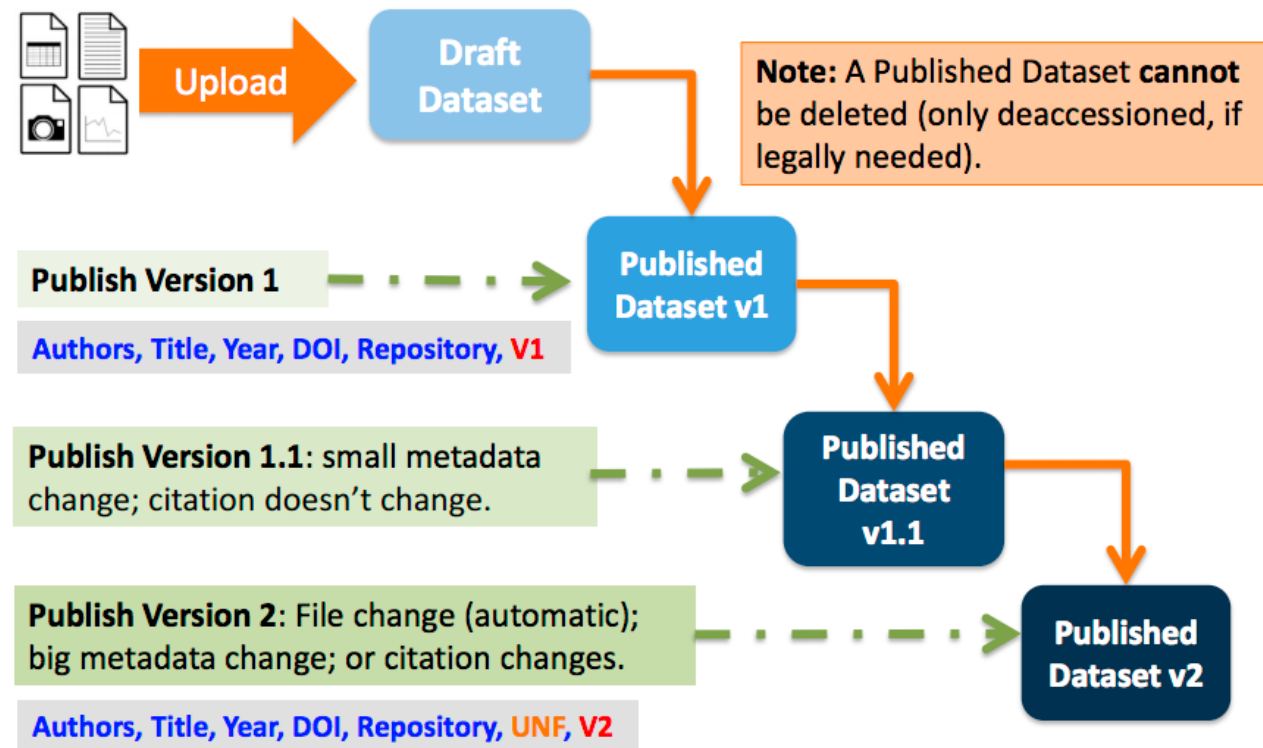
<https://cran.r-project.org/web/packages/DDIwR/index.html>

Semantic Versioning Convention

Currently supported on Dataverse:

major revision / minor revision

BMF/v1.12



R Package for Dataverse – depositing a dataset:

```
# create the dataset. e/g/  
ds <- create_dataset("mydataverse") # pick a name of dataset  
  
# add files  
tmp <- tempfile() # In this example, we write to a temporary destination  
write.csv(iris, file = tmp)  
add_dataset_file(file = tmp, dataset = ds)  
  
# publish dataset  
publish_dataset(ds)  
  
# dataset will now be published  
get_dataverse("mydataverse")
```

```
library("dataverse")  
Sys.setenv("DATAVERSE_SERVER" = "dataverse.harvard.edu")  
Sys.setenv("DATAVERSE_KEY" = "examplekey12345")  
create_dataset()    # dataset can be multiple tables  
add_dataset_file()  # add new table, dictionary, crosswalk, code  
update_dataset()    # update description, documentation  
publish_dataset()   # can chose MINOR or MAJOR revision
```

Planned Support for Data Provenance (workflows):

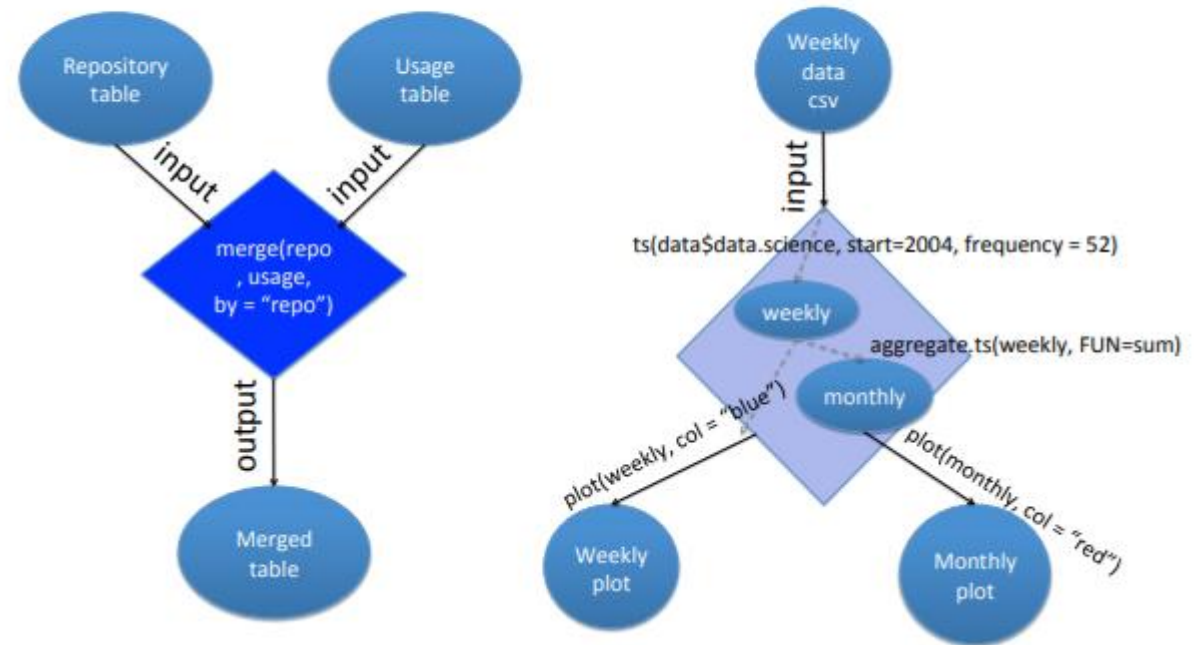
What we propose

- Describe Provenance with **code** used to transform or merge datasets
- Represent **Provenance as a graph**, connecting the code with the input and output datasets (using DOIs)
- Include Provenance graph in Dataset **metadata**

Anticipated Collaborations:

- With DataCite, enhance their metadata schema to support provenance graph
- With USENIX, integrate their open access repository with the citation + provenance work from this project

Provenance examples with R Code



Integrating Data Citation with Provenance

Margo Seltzer, Mercè Crosas, Gary King



HARVARD
School of Engineering
and Applied Sciences



What is Dataverse?

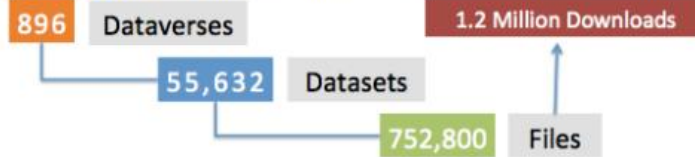
Software framework for publishing, citing and preserving research data
(open source on [github](https://github.com) for others to install)

Provides incentives for researchers to share:

- Recognition & credit via **data citations**
- Control over data & branding
- Fulfill journal data availability and funder data management requirements



Harvard Dataverse (open to all; repository instance at Harvard) currently has:



Data Citation with Dataverse

Integrates with DataCite service for minting persistent identifiers (DOIs) and registering dataset citation metadata

Authors, Year, Dataset Title, DOI, Data Repository, UNF, version

Attribution to data
authors and distributors

Fingerprint (UNF) to verify
dataset, and version to
specify what data are being
referenced. UNF does **not**
depend on the data format.

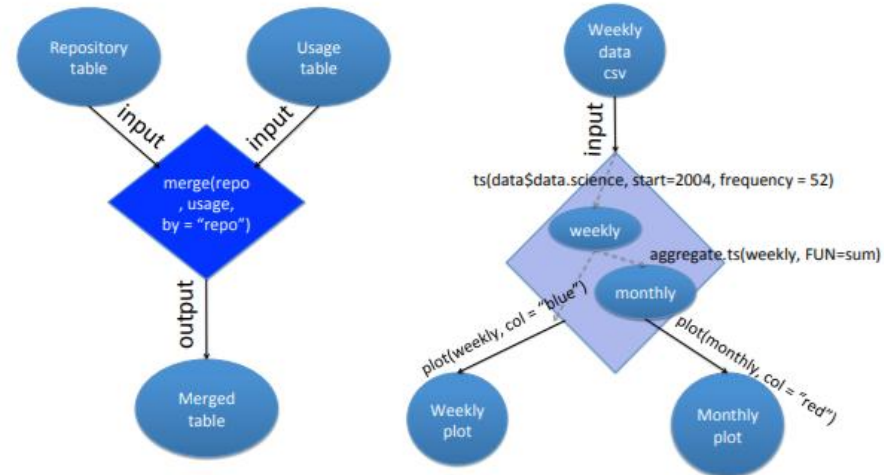
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Provenance examples with R Code



Dataverse [Q](#) [About](#) [Guides](#) [Support](#) [Sign Up](#) [Log In](#)

[Tom Brady Dataverse \(New England Patriots\)](#)

[Root Dataverse](#) > [Tom Brady Dataverse](#) > [Free Samples](#) > **50by1000.tab**

Metrics
0 Downloads

   [Download](#)

50by1000.tab

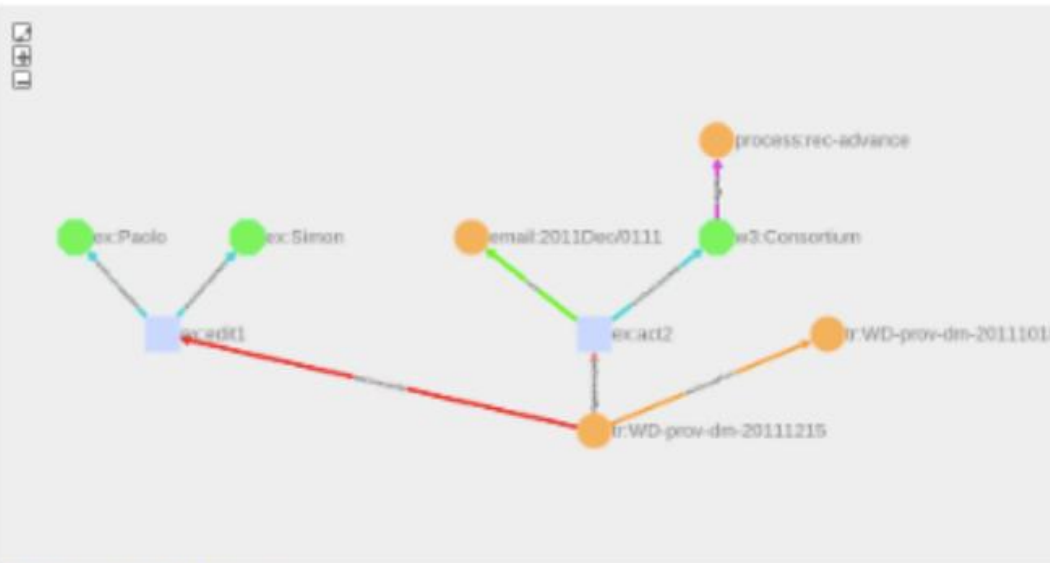
Brady, Tom, 2016, "Free Samples",
doi:10.5072/FK2/FVXYPF, Root Dataverse, V1,
UNF:6:5D4H5Hw-eMupH538Oz2w==; 50by1000.tab
[FileName], UNF:6:x10r+Q0EK6aF8M-eKzGe== [fileUNF]

[Learn about Data Citation Standards.](#)

Tabular Data - 102.5 KB - Last Updated: Oct 25, 2016
50 Variables, 1000 Observations - UNF:6:x10r+Q0EK6aF8M-eKzGe==

[Data](#) [Playbook](#) [Geospatial](#) [Time Series](#)

[Metadata](#) [Provenance](#) [Versions](#)



[Show Ancestors](#) [Show Successors](#)

Data File Transformations:

- Use Core Provenance Library (CPL)
- Use W3C PROV schema
- Display as Provenance Visualizations
- Currently in development for a future Dataverse release

<https://dataverse.org/files/dataverseorg/files/metadataprovenance-mercecrosas.pdf>